MILITARY SPECIFICATION

DETONATOR, STAB, M46
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading assembly and packing for one type of stab detonator designated as Detonator Stab, M46.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-A-2550 — Ammunition and Special Weapons, General Specifications for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-650 — Explosive: Sampling, Inspection and Testing.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548099 — Box, Packing, Ammunition for Detonators.

7548103 — Box, Set-up, Packing, Ammunition, Outer for Boosters, Detonators, or Primers, Assembly, Details, Packing and Marking.

7548104 — Box, Set-up, Packing, Ammunition, Inner, for Boosters, Deto-

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nators and Primers, Assembly, Details, Packing and Marking.

8797793 — Detonator, Stab, M46, Loading Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8797793 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator. The detonator shall comply with all requirements specified on Drawing (dwg.) 8797793 and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide. The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent when tested as specified in 4.3.2.

3.4 Moisture content of primer mixture. The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent when tested as specified in 4.3.3.

3.5 Functioning.

3.5.1 *Sensitivity.* The detonator shall function on the first attempt and shall cause perforation of a 0.1345 inch thick lead disc, to a diameter of not less than 0.3125 inch when tested as specified in 4.3.5.1.

3.5.2 *Waterproofness.* The detonator shall comply with the requirements of 3.5.1 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 70° F plus or minus 10°, when tested as specified in 4.3.5.2.

3.6 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

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(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Statement of findings on the primer mixture and associated ingredients and a certificate of conformance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of detonators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL–STD–105 (or Standard MIL–STD–1235, when applicable). Each lot shall contain:

(a) Parts of one lot interfix number from one manufacturer.

(b) Lead azide from not more than one lot from one manufacturer.

(c) Primer mixture from one batch or blend of batches which results in homogeneity of all primer mixture used in the lot.

(d) RDX from not more than one lot from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used, for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Assembly, prior to loading and closing (see dwg. 8797793).

Categories	Defects	Method of inspection	Code No. (see 6.2)
Critical:			
1.	Interior coating incomplete or inadequate	Visual	01001
Major: None defined			
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	01002

4.2.2.2 Detonator assembly (see dwg 8797793).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Cup split or cracked	Visual	02001
Major:	AQL 1.00 percent		
101.	Total length	Gage	02002
102.	Diameter	Gage	02003
103.	Disc loose	Manual	02004
104.	Color of disc improper	Visual	02005
105.	Not crimped full 360°	Visual	02006
106.	Disc punctured or torn	Visual	02007
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02008

4.2.2.3 Box, set-up, inner (see dwg. 7548104).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Box damaged	Visual	03001
102.	Number of detonators incorrect	Visual	03002
Minor:	AQL 1.00 percent		
201.	Marking misleading or unidentifiable	Visual	03003
202.	Any packing material missing	Visual	03004

4.2.2.4 Box, set-up, outer (see dwg. 7548103).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Number of inner set-up boxes incorrect	Visual	04001
102.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal the box completely	Visual	04002
Minor:	AQL 1.50 percent		
201.	Box torn, cut or punctured	Visual	04003
202.	Marking misleading or unidentifiable	Visual	04004
203.	Contents move when box is shaken	Manual	04005

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4.2.2.5 Sealed barrier bag (see dwg. 7548103).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Seal missing incomplete or incorrectly located	Visual	05001
102.	Bag puntured, torn or cut	Visual	05002
Minor: None defined			

4.2.2.6 Box, packing, ammunition (see dwg. 7548099).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Board broken or split	Visual	06001
102.	Strapping missing, broken or loose	Visual	06002
Minor:	AQL 1.50 percent		
201.	Marking misleading or unidentifiable	Visual	06003
202.	Strapping improperly assembled	Visual	06004
203.	Contents move when box is tilted	Manual	06005

4.2.3 *Testing.*

4.2.3.1 *Moisture content of lead azide (see 3.3)—Major defect—Code No. 07001.* The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample in sufficient quantity to perform the test detailed in 4.3.1 shall be selected and tested as specified in 4.3.1. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of lead azide shall be rejected. If loading has begun, action on the unloaded azide shall be as specified above and all detonators loaded with questionable lead azide shall be rejected.

4.2.3.2 *Binder-lubricant content of RDX (see dwg. 8797793) — Major defect — Code No. 07002.* The total binder-lubricant, calcium resinate and graphite content of the RDX shall be determined for each lot at the time of loading. A sample of 2 ounces of the RDX and binder-lubricant mixture shall be selected and tested as specified in 4.3.2. If the total binder-lubricant, calcium resinate or graphite is in excess of the percentages specified on the applicable drawing and if loading has not begun, the lot of RDX shall be rejected until proper corrective action has been accomplished as verified by performing the test for binder-lubricant content as specified in 4.3.2. If loading has begun, action on the unloaded RDX shall be as specified above and all detonators loaded with the questionable RDX shall be selected.

4.2.3.3 *Primer mixture*

4.2.3.3.1 *Moisture content (see 3.4)—Major defect—Code No. 08001.* The moisture content of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample, in sufficient quantity to perform the test detailed in 4.3.3 shall be selected and tested as specified in 4.3.3. If the moisture content is in excess of the percentage specified and if loading has not begun, the batch or batches of the primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.2.3.3.2 *Composition (see dwg. 8797793)*

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—*Major defect—Code No. 08002.* The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample in sufficient quantity to perform the test detailed in 4.3.4, shall be selected and tested as specified in 4.3.4. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing and if loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, detonators loaded with the questionable mixture shall be rejected.

4.2.3.4 Functioning

4.2.3.4.1 *Sensitivity (see 3.5.1) — Major defect—Code No. 09001.* A sample of 750 detonators shall be selected at random from each lot. If 5 or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.3.5.1 using test equipment as specified in 4.2.4.

4.2.3.4.2 *Waterproofness (see 3.5.2)—Major defect—Code No. 09002.* A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.3.5.2 using test equipment as specified in 4.2.4.

4.2.3.5 *Heat seal test of sealed bag (see dwg. 7548103) — Major defect — Code No. 09003.* Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL–P–116.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8797793 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Moisture content of lead azide.* The moisture content of the lead azide shall be determined in accordance with MIL–STD–650 Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluorothylene.

4.3.2 *Binder-lubricant content of RDX.* The calcium resinate, graphite, and total binder-lubricant content of the RDX to be used in the detonator shall be determined as follows:

4.3.2.1 *Calcium resinate.* An accurately weighed portion of 5 grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliter (ml.) portions of chloroform which have been previously saturated with RDX, main-

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taining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20 ml. portion of chloroform to insure complete removal of the calcium resinate. The residue shall be aspirated until the odor of chloroform is no longer detectable; the crucible and contents shall be placed in a drying oven at 95–105 C. for ½ hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying, and weighing with individual 20 ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the crucible plus contents shall be calculated to percent calcium resinate in the sample on a moisture-free basis as follows:

$$\text{Percentage of Calcium Resinate} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible plus contents.

W = weight of sample.

4.3.2.2 *Graphite.* The crucible and contents remaining after the removal of calcium resinate shall be washed with a minimum of three 20 ml. portions of acetone, each portion maintained in contact with the residue for approximately 1 minute with continuous stirring before applying suction, until the subsequent test for RDX in the acetone washings is negative. The crucible and contents shall be washed with a minimum of 3 ml. portions of acetone, each portion maintained in contact with the residue for approximately 1 minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25 ml. beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 95–105° C. for ½ hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture free basis as follows:

$$\text{Percentage of Graphite} = \frac{100A}{W}$$

where:

A = increase in weight of filtering crucible.

W = weight of sample.

4.3.2.3 *Total binder-lubricant.* The percentage of total binder-lubricant in the sample shall be calculated by adding the percentage of calcium resinate and graphite found.

4.3.3 *Moisture content of primer mixture.* The moisture content of the primer mixture shall be determined in accordance with MIL-STD-650 method 101.5 except, use a one gram sample.

4.3.4 *Composition of primer mixture.* The composition of the primer mixture shall be determined by performing the following tests in the sequence specified:

4.3.4.1 *Barium nitrate.* An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml. capacity. Three ml. of lead azide saturated distilled water shall be added at a temperature of 50° C. plus or minus 2° C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60°

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C. plus or minus 5° C. for 30 minutes, cooled in a disiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{M(A-B)}{1-.01M}$$

where:

A = weight of crucible and sample.

B = weight of the crucible and sample after extractions with lead azide saturated water.

M = moisture content of sample obtained from 4.3.3 in percent.

4.3.4.2 *Lead styphnate*

4.3.4.2.1 *Basic lead styphnate (when applicable)*. The residue obtained in the determination of barium nitrate shall be extracted with six-ml. portions of saturated ammonium acetate solution at about 25° C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml. aliquot of this solution shall be transferred to a 50 ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this selection shall be transferred to a corex glass spectrophotometric cell (see 6.5) having a width of approximately 1 centimeter (cm.) and its absorbance density determined at a wave length of 410 millimicrons. As reference, a similar cell containing acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F\ (A-B)}{W\ (1-.01M)\ D}$$

where:

A = optical density of basic lead styphnate solution

B = optical density of ammonium acetate solution.

D = width of corex cell, cm.

M = moisture content of sample obtained from 4.3.3 in percent.

W = weight of sample.

F = Factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

4.3.4.2.2 *Determination of normal lead styphnate (when applicable)*. Extract the residue obtained in the barium nitrate determination with sufficient 5 ml. portions of ammonium acetate solution (saturated) at about 25° C. until the washings are colorless. Agitate each portion for about 1 minute, then allow each portion to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracecene determination. Transfer the filtrate and washings quantitatively to a 250 ml. calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated

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pipette, transfer a 2 ml. aliquot of this solution to a second 100 ml. calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of approximately 1 cm. and determine its absorbance density at a wave length of 410 millimicrons. Use as a reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wavelength of 410 millimicrons, the absorbance density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

$$\text{Normal lead styphnate, percent} = \frac{F\ (A-B)}{WD}$$

where:

A = optical density of normal lead styphnate solution.

B = optical density of ammonium acetate solution.

D = width of corex cell, cm.

W = weight of the dried sample.

F = Factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

4.3.4.3 *Tetracene.* The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained ot 60° C. plus or minus 5° C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60° C. plus or minus 5° C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D\ 100}{W\ (1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.

D = weight of crucible and residue after treating with hot water.

W = weight of the sample.

M = moisture content of sample obtained from 4.3.3 in percent.

4.3.4.4 *Antimony sulfide.* The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W\ (1-.01M)}$$

where:

D = weight of crucible and sample after treating with hot water.

E = weight of crucible.

W = weight of the sample.

M = moisture content of sample obtained from 4.3.3 in percent.

4.3.4.5 *Lead Azide.* The lead azide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

4.3.5 *Functioning.*

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4.3.5.1 *Sensitivity.* The detonator shall be assembled to the applicable test fixture specified in 4.2.4, so that the firing pin will strike the sensitive end of the detonator. A steel ball, approximately 15/16 inch in diameter and weighing 1.94 ounces maximum, shall be dropped on the firing pin, to function the detonator, from a height of 2 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.3.5.2 *Waterproofness.* The detonator shall be immersed under water as specified for the time specified. Within one hour after removal from the water the detonator shall be subjected to the sensitivity test specified in 4.3.5.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* Packaging shall be as specified on dwg. 7548104.

5.1.2 *Intermediate packaging.* The primers shall be packaged in accordance with dwg. 7548103.

5.2 Packing.

5.2.1 *Level C.* The containers shall be packed in accordance with dwg. 7548099.

5.3 Marking. Inner, outer and exterior containers shall be marked in accordance with dwgs. 7548103, 7548104, 7548099.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608

CDCF — Contractor's design responsibility on contractor format.

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AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and nonmandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspertion equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completey fabricate, caibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the details and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be nonreproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month

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after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Check test for possible deterioration. The following test is designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in that contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration.

6.4.1 *Sensitivity.* A sample of 750 detonators shall be selected at random from each lot. If 7 or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.3.5.1 using test equipment as specified in 4.2.4.

6.5 A Quartz Ultraviolet Spectrophotometer described in the Journal of American Optical Society, Vol 30 page 683 (1941) was found to be satisfactory for determining the optical density of the styphnate solution.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1375–A892

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Army-Ft Monmouth, NJ-MON 94-64

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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